

Modeling Phosphorylation Site Information in WikiPathways

This document describes how to use public databases to find relevant phosphorylation site information in the context of a pathway, for the purpose of annotating pathway nodes. The information is entered as structured comment in PathVisio, which can then be utilized data visualization in tools like [Cytoscape](#).

Structured comment

The format of the structured comment is shown below:

```
parentid=Q92934; parentsymbol=BAD; site=PFrGrSRSAPPNLWA;  
position=ser99; sitegrpId=447862; ptm=p; direction=d
```

parentid: UniProtKB identifier for the parent protein

parentsymbol: Gene/protein symbol for the parent protein

site: the protein sequence flanking the phosphorylation site, 7 aa in each direction

position: the phosphorylated amino acid and its numeric position

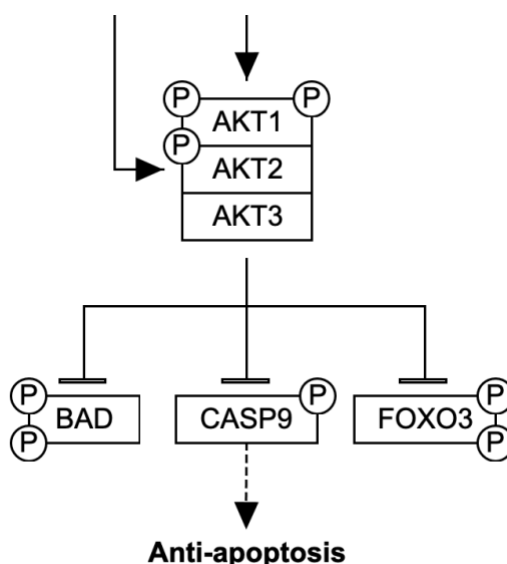
sitegrpId: the Site Group ID from the [PhosphoSitePlus database](#)

ptm: the type of post-translational modification; in the case of phosphorylation encoded as “p”

direction: describes the effect that the phosphorylation has on the parent protein, “d” for down and “u” for up

Step-by-step instructions

The instructions use an example phosphorylation event from a pathway; the phosphorylation of the BAD protein by AKT1/2/3, which causes inactivation of BAD, ultimately leading to an anti-apoptotic effect.



1. Navigate to PhosphoSitePlus (<https://www.phosphosite.org/homeAction>) and enter the protein name, BAD, in the search field at the top right. In the list of results, select the result for BAD, with gene name BAD.
2. On the **Protein Information** tab, find the:
 - a. parentid: Q92934 (UniProtKB ID)
 - b. parentsymbol: BAD (Gene Symbol)
3. In the Upstream tab, look for Akt1/2/3 listed under Kinase, in vitro or Putative in vivo kinase. In this case, S99-p is listed as one such site.
4. Click on the S99-p site to open the site-specific page. Find the
 - a. position: Ser99
 - b. sitegroupid: 447862 (Site Group ID)
 - c. site: PFrGrsRsAPPNLWA
5. Next, confirm that the **Effects of modification on biological processes** makes sense in the context of the pathway. For S99-p, it states “apoptosis, altered” and “apoptosis, inhibited”, which both match the pathway context.
6. The direction parameter describes the effect that the phosphorylation has on the parent protein (BAD). In this case, the “apoptosis, inhibited” is pretty clear that the effect of the phosphorylation is an inhibition, or “down”, so we will record this as “d” for direction. If the direction of the effect is not clear from PhosphoSitePlus, we use the direction indicated in the pathway model. For example, an inhibition interaction between AKT1/2/3 and BAD indicates down-regulation.
7. Construct the custom comment using the recorded information:

```
parentid=Q92934; parentsymbol=BAD; site=PFrGrsRsAPPNLWA;  
position=ser99; sitegroupid=447862; ptm=p; direction=d
```

Home > Phosphorylation Site Page: > Ser99 - BAD (human) ←

Site Information

PFrGrSRsAPPNLWA [SwissProt](#) [Entrez-Gene](#) ←
Blast this site against: [NCBI](#) [SwissProt](#) [PDB](#)
Site Group ID: 447862 ←
Associated spectra: 2 CST

In vivo Characterization

Methods used to characterize site *in vivo*: [32P] bio-synthetic labeling (289), immunoassay (2 , 81), immunoprecipitation (68 , 77 , 294 , 296 , 299), mass spectrometry (9 , 10 , 11 , 13 , 14 , 15 , 17 , 19 , 20 , 21 , 22 , [view more >](#)

Disease tissue studied: bone cancer (36), brain cancer (68), glioblastoma (68), glioblastoma multiforme (68), glioma (68), breast cancer (2 , 5 , 16 , 20 , 21 , 30 , 31 , 227 , 249 , 273 , 276 , 282), [view more >](#)

Relevant cell line - cell type - tissue: 'neuron, cerebellar granule'-brain (299), 'pancreatic, ductal'-pancreas (24), 'stem, embryonic' (172), 293 (epithelial) [AT1 (human), transfection, AT1R stable transfected [view more >](#)

Upstream Regulation

Regulatory protein: 14-3-3 epsilon (human) (7), Ack (human) (3), Akt1 (human) (33 , 36 , 297), CDC25A (human) (7), CDC42 iso1 (human) (269), FLT3 (human) (237), GRB10 (human) (249), LAPTM4B (human) (18), LKB1 (human) (228), PAK1 (human) (269), PDK1 (human) (68), Pim1 (human) (191), Pim3 (human) (272), PINK1 (human) (79), PKCA (human) (16 , 295), PKCE (human) (295), PKCT (human) (295), PKCZ (human) (68), PKN2 (human) (296), PPP1CC (human) (78), PTEN (human) (4), RAC1 (human) (269), RAF1 (human) (249 , 269), RMP (human) (78), SHIP (human) (279), SLC1A5 (human) (251), TSC1 (human) (268), TSC2 (human) (268) [view less <](#)

Putative in vivo kinases: Akt1 (human) (3 , 148), BRAF (human) (148), PAK1 (human) (148 , 298), PKACA (human) (148), PKCI (human) (68 , 277), RAF1 (human) (148)

Kinases, *in vitro*: Akt1 (human) (77 , 148 , 294 , 299), ARAF (human) (148), BRAF (human) (148), PAK1 (human) (148 , 298), PKCE (human) (144), PKCI (human) (68 , 277), PKCT [view more >](#)

Treatments: 15-epi-LXA4 (250), 2-(3-(methylamino)phenyl)-6-(pyrrolidin-1-yl)quinolin-4-one (76), 2-deoxyglucose (165), 4-HT (227), AG1478 (79 , 269), Akt_inhibitor_VIII (79), [view more >](#)

Downstream Regulation

Effects of modification on BAD: activity, induced (286 , 287), molecular association, regulation (3 , 68 , 77 , 85 , 148 , 298), phosphorylation (77)

Effects of modification on biological processes: apoptosis, altered (106 , 228 , 282 , 297), apoptosis, inhibited (2 , 3 , 4 , 6 , 7 , 65 , 286 , 296 , 298), carcinogenesis, induced (2 , 3 , 4), cell cycle regulation (2), cell growth, altered (286), cell motility, induced (2), transcription, inhibited (247) [view less <](#)

Induce interaction with: 14-3-3 beta (human) (68 , 85 , 298), 14-3-3 zeta (human) (148)

Inhibit interaction with: 14-3-3 beta (human) (77), Bcl-2 (human) (3 , 85 , 298), Bcl-xL (human) (3 , 68 , 298)

Disease / Diagnostics Relevance

Relevant diseases: breast adenocarcinoma (16), breast cancer, triple negative (2), HNSCC (81), ovarian cancer (65)

Note: If the information available from PhosphoSitePlus is not conclusive or clear, simply skip that site.